

Example 1

- Every cadet runs a timed sprint course as part of camp trials.
- The coaches want us to determine if the Thunder Squadron and Lightning Squadron are finishing the sprint course at different times.
- Using the `independent_means()` function,

```
wonderbolt_camp |> independent_means(outcome = sprint_time_sec,  
                                     grouping = squadron,  
                                     alternative = "two",  
                                     mu = 0,  
                                     alpha = 0.05)
```

Independent means (Welch's t-test)

$\bar{x}$ [Lightning Squadron] = 44.4643

$\bar{x}$ [Thunder Squadron] = 41.8188

Point estimate: $\bar{x}$ [Lightning Squadron] - $\bar{x}$ [Thunder Squadron] = 2.6455

95% CI for μ [Lightning Squadron] - μ [Thunder Squadron]: (0.3719, 4.9192)

Hypotheses:

$H_0: \mu$ [Lightning Squadron] - μ [Thunder Squadron] = 0

$H_1: \mu$ [Lightning Squadron] - μ [Thunder Squadron] $\neq$ 0

Test statistic: $t(27.81) = 2.384$

p-value: 0.024

Conclusion: Reject the null hypothesis ($p = 0.024 < \alpha = 0.050$)

- **Hypotheses:**

- **Test Statistic and p -Value**

- **Rejection Region**

- **Conclusion and interpretation**

Example 2

- At camp, judges score each cadet's aerial loop maneuver on a 0–100 accuracy scale during trials.
- Based on her gut feeling, the head judge believes that, on average, the Thunder Squadron is performing more accurate loop maneuvers.
- Using the `independent_means()` function,

```
wonderbolt_camp |> independent_means(outcome = loop_accuracy_pct,  
                                     grouping = squadron,  
                                     alternative = "less",  
                                     mu = 0,  
                                     alpha = 0.05)
```

Independent means (Welch's t-test)

$\bar{x}$ [Lightning Squadron] = 78.4071

$\bar{x}$ [Thunder Squadron] = 81.2875

Point estimate: $\bar{x}$ [Lightning Squadron] – $\bar{x}$ [Thunder Squadron] = -2.8804

95% CI for μ [Lightning Squadron] – μ [Thunder Squadron]: (-7.606, 1.8453)

Hypotheses:

$H_0: \mu$ [Lightning Squadron] – μ [Thunder Squadron] ≥ 0

$H_1: \mu$ [Lightning Squadron] – μ [Thunder Squadron] < 0

Test statistic: $t(25.18) = -1.255$

p-value: 0.111

Conclusion: Fail to reject the null hypothesis ($p = 0.111 \geq \alpha = 0.050$)

- **Hypotheses:**

- **Test Statistic and p -Value**

- **Rejection Region**

- **Conclusion and interpretation**

Example 3

- Each cadet's aerial accuracy is scored when they arrive at camp, and scored again at the end after two weeks of drills.
- Coaches want to know whether cadets are actually flying more accurately by the end of camp.
- Using the `dependent_means()` function,

```
wonderbolt_camp |> dependent_means(col1 = accuracy_score_post,  
                                   col2 = accuracy_score_pre,  
                                   alternative = "greater",  
                                   mu = 0,  
                                   alpha = 0.05)
```

Paired (dependent) means

Complete pairs: 30

Point estimate: $\bar{x}[\text{accuracy_score_post}] - \bar{x}[\text{accuracy_score_pre}] = 6.7267$

SD of differences: 5.1583

95% CI for $\mu[\text{accuracy_score_post}] - \mu[\text{accuracy_score_pre}]$: (4.8005, 8.6528)

Hypotheses:

$H_0: \mu[\text{accuracy_score_post}] - \mu[\text{accuracy_score_pre}] \leq 0$

$H_1: \mu[\text{accuracy_score_post}] - \mu[\text{accuracy_score_pre}] > 0$

Test statistic: $t(29) = 7.143$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- **Hypotheses:**

- **Test Statistic and p -Value**

- **Rejection Region**

- **Conclusion and interpretation**

Example 4

- Each cadet's recovery time after a hard trial is timed at the start of camp and again at the end.
- Coaches want to know whether cadets are recovering faster by the end of camp.
- Using the `dependent_means()` function,

```
wonderbolt_camp |> dependent_means(coll = recovery_time_min_post,  
                                   col2 = recovery_time_min_pre,  
                                   alternative = "less",  
                                   mu = 0,  
                                   alpha = 0.05)
```

Paired (dependent) means

Complete pairs: 30

Point estimate: $\bar{x}[\text{recovery_time_min_post}] - \bar{x}[\text{recovery_time_min_pre}] = -3.26$

SD of differences: 3.3987

95% CI for $\mu[\text{recovery_time_min_post}] - \mu[\text{recovery_time_min_pre}]$: (-4.5291, -1.9909)

Hypotheses:

$H_0: \mu[\text{recovery_time_min_post}] - \mu[\text{recovery_time_min_pre}] \geq 0$

$H_1: \mu[\text{recovery_time_min_post}] - \mu[\text{recovery_time_min_pre}] < 0$

Test statistic: $t(29) = -5.254$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- **Hypotheses:**
- **Test Statistic and p -Value**
- **Rejection Region**
- **Conclusion and interpretation**

Let's Practice! 1

- Spectators toss bits to cadets who pull off a standout run during trials.
- Camp organizers want to know if the squadrons are earning prize bits equally.

```
wonderbolt_camp |> independent_means(outcome = prize_bits_won,  
                                     grouping = squadron,  
                                     alternative = "two",  
                                     mu = 0,  
                                     alpha = 0.05)
```

Independent means (Welch's t-test)

$\bar{x}[\text{Lightning Squadron}] = 27.7857$

$\bar{x}[\text{Thunder Squadron}] = 36.1875$

Point estimate: $\bar{x}[\text{Lightning Squadron}] - \bar{x}[\text{Thunder Squadron}] = -8.4018$

95% CI for $\mu[\text{Lightning Squadron}] - \mu[\text{Thunder Squadron}]$: (-27.9469, 11.1433)

Hypotheses:

$H_0: \mu[\text{Lightning Squadron}] - \mu[\text{Thunder Squadron}] = 0$

$H_1: \mu[\text{Lightning Squadron}] - \mu[\text{Thunder Squadron}] \neq 0$

Test statistic: $t(26.86) = -0.882$

p-value: 0.385

Conclusion: Fail to reject the null hypothesis ($p = 0.385 \geq \alpha = 0.050$)

- **Hypotheses:**

- **Test Statistic and p -Value**

- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 2

- Spectators tip cadets with bits after their runs. The amount tipped is recorded once early in camp and once again near the end.
- Camp organizers want to know whether cadets are earning more in tips as they improve over camp.
- Using the `dependent_means()` function,

```
wonderbolt_camp |> dependent_means(coll = tip_bits_post,  
                                   col2 = tip_bits_pre,  
                                   alternative = "greater",  
                                   mu = 0,  
                                   alpha = 0.05)
```

Paired (dependent) means

Complete pairs: 30

Point estimate: $\bar{x}[\text{tip_bits_post}] - \bar{x}[\text{tip_bits_pre}] = 17.9333$

SD of differences: 13.4982

95% CI for $\mu[\text{tip_bits_post}] - \mu[\text{tip_bits_pre}]$: (12.893, 22.9737)

Hypotheses:

$H_0: \mu[\text{tip_bits_post}] - \mu[\text{tip_bits_pre}] \leq 0$

$H_1: \mu[\text{tip_bits_post}] - \mu[\text{tip_bits_pre}] > 0$

Test statistic: $t(29) = 7.277$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- **Hypotheses:**

- **Test Statistic and p -Value**

- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 3

- A sound crew records the peak crowd noise (in decibels) during each cadet's run.
- Camp organizers believe that the Thunder Squadron is inspiring a higher level of noise.
- Using the `independent_means()` function,

```
wonderbolt_camp |> independent_means(outcome = cheer_decibels,  
                                     grouping = squadron,  
                                     alternative = "less",  
                                     mu = 0,  
                                     alpha = 0.05)
```

Independent means (Welch's t-test)

$\bar{x}[\text{Lightning Squadron}] = 84.9$

$\bar{x}[\text{Thunder Squadron}] = 90.8187$

Point estimate: $\bar{x}[\text{Lightning Squadron}] - \bar{x}[\text{Thunder Squadron}] = -5.9187$

95% CI for $\mu[\text{Lightning Squadron}] - \mu[\text{Thunder Squadron}]$: (-8.6025, -3.235)

Hypotheses:

$H_0: \mu[\text{Lightning Squadron}] - \mu[\text{Thunder Squadron}] \geq 0$

$H_1: \mu[\text{Lightning Squadron}] - \mu[\text{Thunder Squadron}] < 0$

Test statistic: $t(21.92) = -4.575$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- **Hypotheses:**

- **Test Statistic and p -Value**

- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 4

- The camp medic tracks how many days each cadet spends recovering from an injury over the two-week camp.
- Coaches want to know if the two training styles lead to the same number of injury days.
- Using the `independent_means()` function,

```
wonderbolt_camp |> independent_means(outcome = injury_days,  
                                     grouping = squadron,  
                                     alternative = "two",  
                                     mu = 0,  
                                     alpha = 0.05)
```

Independent means (Welch's t-test)

$\bar{x}[\text{Lightning Squadron}] = 1.7143$

$\bar{x}[\text{Thunder Squadron}] = 1.3125$

Point estimate: $\bar{x}[\text{Lightning Squadron}] - \bar{x}[\text{Thunder Squadron}] = 0.4018$

95% CI for $\mu[\text{Lightning Squadron}] - \mu[\text{Thunder Squadron}]$: (-0.515, 1.3186)

Hypotheses:

$H_0: \mu[\text{Lightning Squadron}] - \mu[\text{Thunder Squadron}] = 0$

$H_1: \mu[\text{Lightning Squadron}] - \mu[\text{Thunder Squadron}] \neq 0$

Test statistic: $t(25.11) = 0.902$

p-value: 0.375

Conclusion: Fail to reject the null hypothesis ($p = 0.375 \geq \alpha = 0.050$)

- **Hypotheses:**

- **Test Statistic and p -Value**

- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 5

- Each cadet's wing-flap rate while hovering in place is measured on arrival, then measured again after two weeks of drills.
- Coaches want to know whether training actually changes a cadet's wing-flap rate.
- Using the `dependent_means()` function,

```
wonderbolt_camp |> dependent_means(col1 = wing_flap_rate_post,  
                                   col2 = wing_flap_rate_pre,  
                                   alternative = "greater",  
                                   mu = 0,  
                                   alpha = 0.05)
```

Paired (dependent) means

Complete pairs: 30

Point estimate: $\bar{x}[\text{wing_flap_rate_post}] - \bar{x}[\text{wing_flap_rate_pre}] = 3.7833$

SD of differences: 4.7192

95% CI for $\mu[\text{wing_flap_rate_post}] - \mu[\text{wing_flap_rate_pre}]$: (2.0212, 5.5455)

Hypotheses:

$H_0: \mu[\text{wing_flap_rate_post}] - \mu[\text{wing_flap_rate_pre}] \leq 0$

$H_1: \mu[\text{wing_flap_rate_post}] - \mu[\text{wing_flap_rate_pre}] > 0$

Test statistic: $t(29) = 4.391$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- **Hypotheses:**

- **Test Statistic and p -Value**

- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 6

- Each cadet rates how confident they feel about their flying on a self-reported scale, once at the start of camp and once at the end.
- Camp staff want to know whether two weeks of training changes how confident cadets feel.
- Using the `dependent_means()` function,

```
wonderbolt_camp |> dependent_means(col1 = confidence_rating_post,  
                                   col2 = confidence_rating_pre,  
                                   alternative = "greater",  
                                   mu = 0,  
                                   alpha = 0.05)
```

Paired (dependent) means

Complete pairs: 30

Point estimate: $\bar{x}[\text{confidence_rating_post}] - \bar{x}[\text{confidence_rating_pre}] = 1.0667$

SD of differences: 1.3629

95% CI for $\mu[\text{confidence_rating_post}] - \mu[\text{confidence_rating_pre}]$: (0.5578, 1.5756)

Hypotheses:

$H_0: \mu[\text{confidence_rating_post}] - \mu[\text{confidence_rating_pre}] \leq 0$

$H_1: \mu[\text{confidence_rating_post}] - \mu[\text{confidence_rating_pre}] > 0$

Test statistic: $t(29) = 4.287$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

- **Hypotheses:**
- **Test Statistic and p -Value**
- **Rejection Region**
- **Conclusion and interpretation**